Solutions to JEE Advanced – 2 | JEE 2022 | Paper-1

PHYSICS

1.(D) Since B moves faster than A, clearly it will cover a greater distance before they meet

So, we can look at the situation as B being three-quarters of the circle, i.e. a distance $\frac{3\pi R}{2}$ behind A initially.

Hence, the time instant when they meet is given by

$$\left(\frac{3v}{2}\right)t = vt + \frac{3\pi R}{2} \quad \Rightarrow \quad t = \frac{3\pi R}{v}$$

2.(D) Let the tension in the wire be T and let the velocity of transverse waves in the wire be c

Then, $c = \sqrt{\frac{T}{\left(\frac{m}{L}\right)}}$

The fundamental frequency of the left segment of the wire,

$$f_1 = \frac{c}{2\left(\frac{L}{3}\right)} = \frac{3c}{2L}$$

And the fundamental frequency of the right segment,

$$f_2 = \frac{c}{2\left(\frac{2L}{3}\right)} = \frac{3c}{4L}$$

The minimum frequency at which both segments can resonate together is the least common multiple of these two fundamental frequencies.

So, $f = \frac{3c}{2L} \quad \Rightarrow \quad f = \frac{3}{2}\sqrt{\frac{T}{mL}} \quad \Rightarrow \quad T = \frac{4mLf^2}{9}$

3.(A) $U(x) = x^2 + 8x - 2 = -18 + (x+4)^2$

Comparing with $U(x) = U_0 + \frac{1}{2}k(x - x_0)^2$, we can deduce that $U_0 = -18$, $k = 2$, and $x_0 = -4$

Therefore, the particle is executing SHM with angular frequency, $\omega = \sqrt{\frac{k}{m}} = 5$ rad/s

And, the mean position of the particle is $x_0 = -4$

So, its equation of motion is $x(t) = -4 + A\sin(5t + \phi)$

Here, A and ϕ depend on the initial conditions, i.e. how the motion of the particle begins.

4.(B) The horizontal component of the velocity of the ball remains unchanged throughout.

So, $u\sqrt{\frac{2h_1}{g}} + u\sqrt{\frac{2h_2}{g}} = L \quad \Rightarrow \quad h_2 = \left(\frac{L}{u}\sqrt{\frac{g}{2}} - \sqrt{h_1}\right)^2$

5.(B) Pressure at a point on the interface between the liquids:

$$P_0 - \frac{2\sigma}{\left(\frac{r}{\cos\theta}\right)} + \rho_1 gh = P_0 + \rho_2 gh \quad \Rightarrow \quad h = \frac{2\sigma\cos\theta}{(\rho_1 - \rho_2)rg}$$

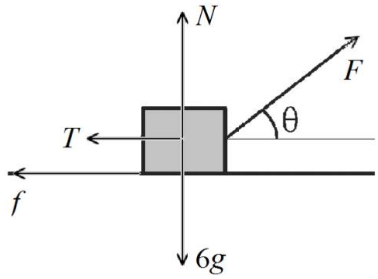
6.(D) From the FBD of B, $T = 1g = 10\ N$

Now, let friction act leftwards on A

Assuming equilibrium,

$$N + F\sin\theta = 6g \qquad \text{and} \qquad T + f = F\cos\theta$$

$$\Rightarrow \qquad N = 60 - 0.6F \qquad \text{and} \qquad f = 0.8F - 10$$



We know that the magnitude of friction must be less than or equal to μN. But, since the friction can act rightwards as well (which can simply be represented in our equations by replacing f by $-f$), the complete inequality on f is

$$-\mu N \le f \le \mu N$$

$$\Rightarrow \qquad -(0.1)(60 - 0.6F) \le (0.8F - 10) \le (0.1)(60 - 0.6F)$$

$$\Rightarrow \qquad -(6 - 0.06F) \le (0.8F - 10) \le (6 - 0.06F)$$

$$\Rightarrow \qquad (0.8F - 10) \ge -(6 - 0.06F) \qquad \text{and} \qquad (0.8F - 10) \le (6 - 0.06F)$$

$$\Rightarrow \qquad 0.74F \ge 4 \qquad \text{and} \qquad 0.86F \le 16$$

$$\Rightarrow \qquad F \ge 5.41 \qquad \text{and} \qquad F \le 18.61$$

So, the complete condition on F is $5.41\ N \le F \le 18.61\ N$

7.(AB) $20\left[C_S(25-20)+150\right] = 88(C_L)(40-25)$

$20(C_S)(25-20) = 8(C_L)(40-25)$

Solving, we get $\qquad C_S = 3.0$ kJ/kg°C $\qquad$ and $\qquad C_L = 2.5$ kJ/kg°C

8.(ABC) Let the initial temperature, pressure and volume on both sides be T_0, P_0 and V_0

The final pressure on both sides will be the same. Let this pressure be P

The final volume on the two sides is $1.1V_0$ and $0.9V_0$

The process on the right side is adiabatic. So,

$$P_0V_0^{5/3} = P(0.9V_0)^{5/3} \qquad \Rightarrow \qquad P = 1.2P_0$$

Let the final temperature on the two sides be T_1 and T_2

Then, $\quad T_1 = \dfrac{(1.2P_0)(1.1V_0)}{P_0V_0}T_0 = 1.32T_0 = 396$ K And, $\quad T_2 = \dfrac{(1.2P_0)(0.9V_0)}{P_0V_0}T_0 = 1.08T_0 = 324$ K

For the right compartment, work done by the gas

$$W_2 = -\Delta U_2 = -\left[\frac{3}{2}(1.2P_0)(0.9V_0) - \frac{3}{2}P_0V_0\right] = -0.12P_0V_0$$

Also, work done by the gas in the left compartment, $W_1 = -W_2 = 0.12P_0V_0$

Change in internal energy of the gas in the left compartment,

$$\Delta U_1 = U_f - U_i = \frac{3}{2}(1.2P_0)(1.1V_0) - \frac{3}{2}P_0V_0 = 0.48P_0V_0$$

So, heat added, $\qquad \Delta Q = \Delta U_1 + W_1 = 0.60P_0V_0$

Hence, $\qquad \Delta Q = (0.60)\left(10^5\right)\left(2\times10^{-3}\right) = 120$ J

9.(ABCD) Let the positive X-axis be towards East and the positive Y-axis towards North

Let the velocity of the train be $\vec{v}_T$

Let the initial velocity of the cyclist be $\vec{v}_{C1} = 4\hat{j}$

Let the initial velocity of the train relative to the cyclist be $\vec{v}_{TC1}$

We are given that $\vec{v}_{TC1} = -12\hat{i} + 12\hat{j}$

Now, we know from the definition of relative velocity that

$$\vec{v}_{TC1} = \vec{v}_T - \vec{v}_{C1}$$

Therefore, $\vec{v}_T - \vec{v}_{C1} = -12\hat{i} + 12\hat{j}$

So, $\vec{v}_T = \vec{v}_{C1} + \left(-12\hat{i} + 12\hat{j}\right) = -12\hat{i} + 16\hat{j}$

Hence, the train's actual velocity is 20 m/s, in a direction $\tan^{-1}\left(\frac{12}{16}\right) = \tan^{-1}\left(\frac{3}{4}\right)$ West of North

Now, if the cyclist moves towards East, his velocity is $\vec{v}_{C2} = 4\hat{i}$

Therefore, the velocity of the train as seen by him now is

$$\vec{v}_{TC2} = \vec{v}_T - \vec{v}_{C2} = \left(-12\hat{i} + 16\hat{j}\right) - 4\hat{i} = -16\hat{i} + 16\hat{j}$$

The velocity of the train as seen by the cyclist is $\vec{v}_{TC} = \vec{v}_T - \vec{v}_C$

So, it is clear that if he orients his velocity opposite to the train's actual velocity, their vector subtraction has the maximum magnitude

10.(BD) Let the diameter of the tube be d, let the length of the water column height be L, let the speed of sound in air be c, and let the tuning fork frequency be f_T

The end correction, $e = 0.3d = 0.015\,\text{m}$

Then, for first resonance, $f_T = \dfrac{c}{4\left((1.2 - L) + e\right)}$

Therefore, $420 = \dfrac{c}{4\left((1.2 - 1.015) + 0.015\right)} \quad \Rightarrow \quad c = 336$ m/s

Now, for second resonance, $f_T = 3\left(\dfrac{c}{4\left((1.2 - L) + e\right)}\right)$

Therefore, for $f_T = 420\,\text{Hz}$ and $c = 336$ m/s, $L = 61.5$ cm

Now, for the n^{th} resonance, $f_T = (2n - 1)\left(\dfrac{c}{4\left((1.2 - L) + e\right)}\right)$

Here, L can take a minimum value of zero. Let us find the value of n corresponding to this maximum value of L.

$$2n - 1 = \frac{4(1.2 - 0 + 0.015)(420)}{336} = 6.075 \qquad \Rightarrow \qquad n \approx 3.50$$

This means that n can take values 1, 2, 3. Therefore, a total of 3 resonances are heard.

Now, with $f_T = 480\,\text{Hz}$, for first resonance,

$$480 = \frac{336}{4\left((1.2 - L) + 0.015\right)} \qquad \Rightarrow \qquad L = 104 \text{ cm}$$

11.(ABD) Let the area of cross-section of the bucket be A_0, and the instantaneous height of the water level in the bucket be h

Then, $A_0 \dfrac{dh}{dt} = Q - \left(\sqrt{2gh}\right)A$

We can see that the rate of increase of h, i.e. $\frac{dh}{dt}$ decreases as h increases. And for some particular value of h, this rate of increases becomes zero. This particular value of h must be the maximum value that h reaches, since it cannot increase any further as the rate of increase has become zero.

Therefore, the maximum value of h is given by: $Q-\left(\sqrt{2gh}\right)A=0$

$$\Rightarrow \qquad h_{\max}=\frac{Q^2}{2gA^2}$$

Now, if this maximum height is less that the height of the bucket,

i.e. $\quad h_{\max}<H \quad \Rightarrow \quad Q<\left(\sqrt{2gH}\right)A$

the water level becomes constant at this value. Otherwise, the bucket overflows.

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The volume of water leaving the hole per unit time $=\left(\sqrt{2gh}\right)A$

So, this quantity increases as long as the level of water in the bucket increases, and becomes constant when the level of water becomes constant.

12.(AD) Torque is the rate of change of angular momentum

Think of torque and angular momentum similar to how you think about force and momentum

During the flight of the particle, the only force acting on it is its weight vertically downward

Clearly, the angular momentum about P is initially zero, and since the torque on the particle (due to the particle's weight) about P is always clockwise, the angular momentum about P keeps increasing

About Q, the angular momentum is anti-clockwise just after projection, and clockwise just before landing. And the torque on the particle (due to the particle's weight) about Q is always clockwise. Hence, we can deduce that the initially anti-clockwise angular momentum about Q must have reduced to zero and then increased in the clockwise direction

13.(200) Net energy exchanged with the surroundings per unit time,

$$H=\sigma A\left(T^4-T_0^4\right)=\left(\frac{40}{7}\times10^{-7}\right)\left(200\times10^{-4}\right)\left(400^4-300^4\right)=200\text{ W}$$

14.(0.75) After the collision, let the velocity of the particle be v towards the right and let the angular velocity of the rod be ω

As the system of rod + particle receives an impulse from the pivot during the collision, we can conserve the angular momentum of the system only about the pivot.

Therefore, $\quad L_{initial}=L_{final} \quad \Rightarrow \quad muL=mvL+\frac{1}{3}mL^2\omega$

From Newton's experimental law (always applied for the point of collision),

$$1(u)=L\omega-v \quad \Rightarrow \quad v=\frac{u}{2} \quad \text{and} \quad \omega=\frac{3u}{2L}$$

So, velocity of the centre of the rod, $\quad v_C=\left(\frac{L}{2}\right)\omega=\frac{3u}{4}$

15.(0.28) Let the tensions in the wires be T_1 and T_2

Balancing horizontal forces,

$$T_1\cos53^\circ=T_2\cos37^\circ \quad \Rightarrow \quad \frac{T_1}{T_2}=\frac{4}{3}$$

Also, if the lengths of the wires are l_1 and l_2, $\dfrac{l_1}{l_2} = \tan 37^{o} = \dfrac{3}{4}$

We know that the extension produced in a wire under tension is

$$\Delta l = \frac{Tl}{YA} \propto \frac{Tl}{Yd^2} \quad 7$$

Therefore, $$\frac{\Delta l_1}{\Delta l_2} = \left(\frac{T_1}{T_2}\right)\left(\frac{l_1}{l_2}\right)\left(\frac{Y_2}{Y_1}\right)\left(\frac{d_2}{d_1}\right)^2 = \left(\frac{4}{3}\right)\left(\frac{3}{4}\right)\left(\frac{1}{8}\right)\left(\frac{3}{2}\right)^2 = \frac{9}{32} = 0.28$$

16.(6.40) Let the speeds of the particles at the given instant be v_1 and v_2

Conserving linear momentum: $mv_1 - 4mv_2 = 0$

Conserving energy: $$-\frac{G(m)(4m)}{a} = -\frac{G(m)(4m)}{\left(\frac{a}{2}\right)} + \frac{1}{2}mv_1^2 + \frac{1}{2}(4m)v_2^2$$

Solving, we get $v_1 = \sqrt{\dfrac{32Gm}{5a}}$ and $v_2 = \sqrt{\dfrac{2Gm}{5a}}$

17.(0.15) The viscous drag force acting on a spherical ball of radius *r* travelling at velocity *v* is given by Stokes Law:

$$F_d = 6\pi\eta rv$$

The buoyant force acting on the sphere when it is completely submerged in the liquid is given by:

$$B = mg\left(\frac{\rho_L}{\rho_S}\right)$$

where ρ_L and ρ_S are the densities of the liquid and the sphere respectively.

The terminal velocity is achieved when the resultant of gravity, buoyant force and viscous drag force becomes zero.

$$F_d + B = mg$$

So, the viscous drag force acting on the sphere after the terminal velocity is achieved is:

$$(F_d)_T = mg - B = mg\left(1 - \frac{\rho_L}{\rho_S}\right)$$

But since the drag force is proportional to the velocity of the sphere, the drag force acting when the velocity is half of the terminal velocity must be half of this force $(F_d)_T$.

So, the required force is $F = \dfrac{(F_d)_T}{2} = \dfrac{mg}{2}\left(1 - \dfrac{\rho_L}{\rho_S}\right) \Rightarrow F = \dfrac{(0.04)(10)}{2}\left(1 - \dfrac{2\times10^3}{8\times10^3}\right) = 0.15\text{ N}$

18.(0.52) Let the initial velocity of the shell be *u*

$$\Rightarrow \quad T = \frac{2u\sin 45^\circ}{g} \qquad \Rightarrow \quad u = \frac{gT}{\sqrt{2}}$$

Let the velocity of the gun immediately afterwards be *v*

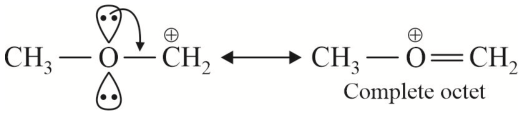
Conserving momentum in the horizontal direction:

$$0 = \left(\frac{M}{25}\right)u\cos 45^\circ + \left(\frac{24M}{25}\right)(-v) \qquad \Rightarrow \qquad v = \frac{u}{24\sqrt{2}} = \frac{gT}{48}$$

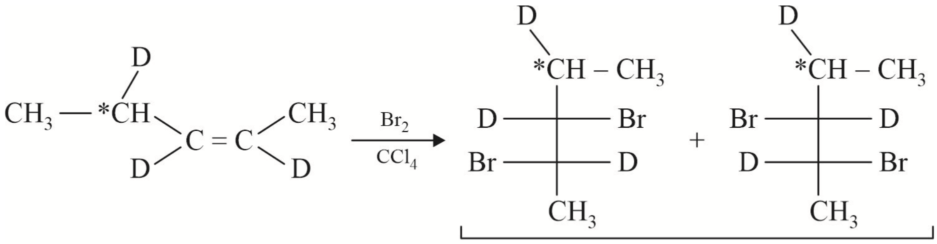
Distance between the gun and the shell at the instant of landing,

$$D = \frac{u^2\sin 90^\circ}{g} + vT = \frac{u^2}{g} + \frac{gT^2}{48} = \frac{gT^2}{2} + \frac{gT^2}{48} = \left(\frac{25}{48}\right)gT^2 \approx 0.52gT^2$$

CHEMISTRY

1.(B) 

(B) is most stable carbocation among given options.

2.(D) 

Diastereomers

3.(D) Radial node is obtained when $\psi_{2s} = 0$

$$\Rightarrow \quad 2 - \frac{r}{a_0} = 0 \quad \Rightarrow \quad 2 = \frac{r}{a_0} \quad \Rightarrow \quad r = 2a_0 = 2 \times 0.53 = 1.06\,\text{Å}$$

4.(C) In low pressure region value of $Z = 1$.

5.(A) On increasing pressure equilibrium shift towards higher density state.

6.(C)

$$X(s) \rightleftharpoons A(g) + 2B(g)$$

Eq. $\quad P_1 \quad 2P_1 + 2P_2$

$$Y(s) \rightleftharpoons 2B(g) + C(g)$$

Eq. $\quad 2P_2 + 2P_1 \quad P_2$

$$P_1(2P_1 + 2P_2)^2 = K_{P_1}$$

$$P_2(2P_1 + 2P_2)^2 = K_{P_2}$$

$$\Rightarrow \quad 4(P_1 + P_2)^3 = K_{P_1} + K_{P_2}$$

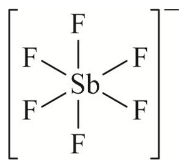
$$P_1 + P_2 = \left(\frac{K_{P_1} + K_{P_2}}{4}\right)^{1/3} = \left(\frac{32}{4} \times 10^{-3}\right)^{1/3} = 2 \times 10^{-1} = 0.2\,\text{atm}$$

$$P_{total} = 3P_1 + 3P_2 = 3 \times 0.2 = 0.6\,\text{atm}$$

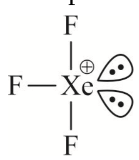
7.(ABC) Correct order of effective nuclear charge

$$\Rightarrow \quad Al < Al^{+} < Al^{2+} < Al^{3+}$$

8.(ABD) SbF_6^- : Octahedral



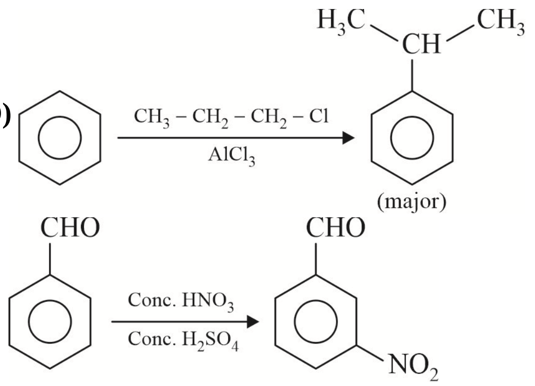
XeF_3^+ : T-shaped



9.(AB) Alkali metal hydrides are mostly ionic and high melting solids.
Mg don't impart characteristic colour to the Bunsen flame.

10.(AB) A and B can act as buffer.

11.(AD)



12.(CD) Given reaction is spontaneous when $T > \dfrac{\Delta H}{\Delta S}$

$$T > \frac{61.17 \times 10^3}{132} \quad \Rightarrow \quad T > 463.4\,K$$

13.(80) $CuS \longrightarrow Cu^{2+} + SO_2$ $\qquad$ $Cu_2S \longrightarrow Cu^{2+} + SO_2$

n.f. = 6 $\qquad$ n.f. = 8

x mol $\qquad$ y mol

$$(eq)_{CuS} + (eq)_{Cu_2S} + (eq)_{Fe^{2+}} = (eq)_{MnO_4^-}$$

$$(x \times 6) + (y \times 8) + \left(\frac{1 \times 1 \times 200}{1000}\right) = \left(\frac{0.4 \times 5 \times 400}{1000}\right)$$

$$6x + 8y = 0.6$$

$$\Rightarrow \quad 3x + 4y = 0.3 \qquad \ldots\ldots\text{(i)}$$

$$96x + 160y = 10$$

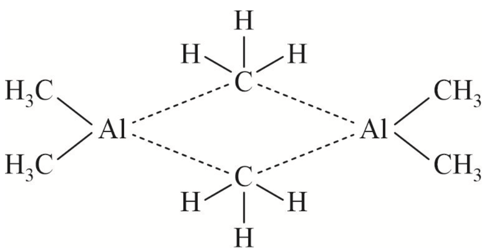
$$\Rightarrow \quad 9.6x + 16y = 1 \qquad \ldots\ldots\text{(ii)}$$

By solving equation (i) and (ii)

$$x = \frac{1}{12},\ y = \frac{1}{80}; \qquad \%\ CuS = \frac{\left(\frac{1}{12} \times 96\right)}{10} \times 100 = 80\%$$

14.(16) There are 4 geometrical isomerism sites and given compound is unsymmetrical.
Total number of stereoisomers $= 2^4 = 16$

15.(24)



Number of $2c - 2e^-$ bond $(Z_1) = 22$

Number of $3c - 2e^-$ bond $(Z_2) = 2 \quad \Rightarrow \quad Z_1 + Z_2 = 24$

16.(6) $H_3Si \leftarrow N = C = O$; 3π bonds $\qquad$ $H_3N \cdot BF_3$: $H_3N \rightarrow BF_3$

HNO_3 : $HO - \overset{\overset{O}{\uparrow}}{N} = O$ $\qquad$ CO : $C \overset{\leftarrow}{\equiv} O$

$X = 3,\ Y = 3$

17.(54) $CH_3 - CH_3 \xrightarrow[hv]{Br_2} \underset{(A)}{CH_3 - CH_2 - Br} \xrightarrow[\Delta]{\text{alc. KOH}} \underset{(B)}{CH_2 = CH_2} \xrightarrow{Br_2/CCl_4} \underset{(C)}{\underset{Br}{CH_2} - \overset{Br}{CH_2}}$

$\xrightarrow[\Delta]{\text{alc. KOH}} \underset{(D)}{\underset{Br}{CH} = CH_2} \xrightarrow[\Delta]{NaNH_2} \underset{(E)}{CH \equiv CH} \xrightarrow{NaNH_2} \underset{(F)}{CH \equiv \overset{\ominus}{C}\overset{\oplus}{Na}} \xrightarrow{CH_3-Br}$

$\underset{(G)}{CH \equiv C - CH_3} \xrightarrow{NaNH_2} \underset{(H)}{\overset{\oplus}{Na}\overset{\ominus}{C} \equiv C - CH_3} \xrightarrow{CH_3Br} \underset{(I)}{CH_3 - C \equiv C - CH_3}$

Molar mass of I is $= 54$

18.(55) Total heat released $= (1 \times 650 \times 20) + (500 \times 4.20 \times 20) = 13000 + 42000 = 55000\,J = 55\,kJ$

MATHEMATICS

1.(C) $ar^5 = 4(ar^3) \Rightarrow r^2 = 4 \Rightarrow r = 2$

$a2^8 - a(2^6) = 192 \Rightarrow a = 1$

$S_n - S_3 = 1016$

$(2^n - 1) - (1+2+4) = 1016$

$2^n = 1024 \Rightarrow n = 10$

2.(A) Let $z = r(\cos\theta + i\sin\theta)$

$$\left|5r\cos\theta + i(5r\sin\theta) + \frac{\cos\theta}{r} - \frac{i\sin\theta}{r}\right| = 3$$

$$\left(5r\cos\theta + \frac{\cos\theta}{r}\right)^2 + \left(5r\sin\theta - \frac{\sin\theta}{r}\right)^2 = 9$$

$$25r^2 + \frac{1}{r^2} + 10\left(\cos^2\theta - \sin^2\theta\right) = 9$$

$$\cos 2\theta = \frac{9 - \left(25r^2 + \frac{1}{r^2}\right)}{10}$$

$25r^2 + \frac{1}{r^2} \geq 10$ **;** $\cos 2\theta = -\frac{1}{10}$

3.(C) $(1) - (2)$

$6x^2 + (0) + (b-c) = 0 \left\langle \begin{matrix}\alpha \\ \beta\end{matrix}\right.$

$\alpha + \beta = 0$

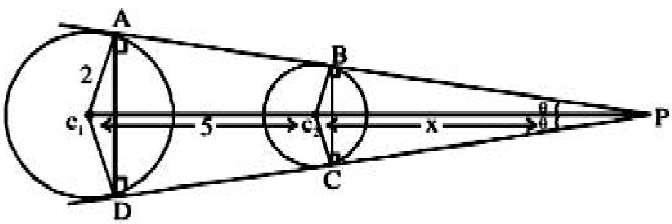
$\alpha + \beta + \gamma = -10$ & $\alpha + \beta + \gamma = -4$

$\gamma = -10$ & $\gamma = -4$

$\delta\gamma = 40$

4.(A) Area $= \frac{S_1^{3/2}}{2a} = \frac{8^{3/2}}{2} = 8\sqrt{2}$

5.(C)



$\frac{x}{x+5} = \frac{1}{2} \Rightarrow x = 5$

$ar(quad\ ABCD) = ar(\Delta PAD) - ar(\Delta PBC) = \frac{1}{2}PA \cdot PD\sin 2\theta - \frac{1}{2}PB \cdot PC\sin 2\theta$

$= \frac{1}{2} \times 96\sin 2\theta - \frac{1}{2} \times 24\sin 2\theta = 36 \times \sin\theta\cos\theta = 72 \times \frac{1}{5} \times \frac{2\sqrt{6}}{5} = \frac{144\sqrt{6}}{25}$

6.(B) $f(-4) = a(-4)^4 + b(-4)^2 - 12 + 7 = 2286$

$f(4) = a(4)^4 + b(4)^2 + 12 + 7 = N$

Subtracting

$N = 2310 = 2^1 \cdot 3^1 \cdot 5^1 \cdot 7^1 \cdot 11^1$

It can be resolved as product of two co-prime divisors $= 2^{n-1} = 2^{5-1} = 16$

Where 'n' is number of distinct primes.

7.(AB) $\cos(\cos x - \sin x) = \cos\left(\frac{\pi}{2} - (\sin x + \cos x)\right)$

$\Rightarrow \quad \cos x - \sin x = 2n\pi \pm \left(\frac{\pi}{2} - (\sin x + \cos x)\right)$

(+) (–)

$\Rightarrow \quad \cos x = n\pi + \frac{\pi}{4} \qquad \sin x = -n\pi + \frac{\pi}{4}$

Clearly, $n = 0$ Clearly $n = 0$

$\Rightarrow \quad \cos x = \frac{\pi}{4} \quad \Rightarrow \quad \sin x = \frac{\pi}{4} \quad \Rightarrow \quad \sin x = \pm\frac{\sqrt{16-\pi^2}}{4}$

8.(CD) $\because \quad z^2 - z = |z|^2 - \frac{64}{|z|^5} =$ a purely real number

$\Rightarrow \quad z^2 - z = \bar{z}^2 - \bar{z} \quad \Rightarrow \quad (z^2 - \bar{z}^2) - (z - \bar{z}) = 0$

If $z - \bar{z} = 0 \quad \Rightarrow \quad z = \bar{z} \quad \Rightarrow \quad z$ is purely real number.

So put $z = x$ in given equation.

$x^2 - x - |x|^2 + \frac{64}{|x|^5} = 0 \quad \Rightarrow \quad x^2 - x - |x|^2 + \frac{64}{x^5} = 0 \quad$ (if $x > 0$)

($\because x < 0$ not possible) $\Rightarrow \quad x^6 = 64 \quad \Rightarrow \quad x = 2 \quad \therefore |z| = 2$

9.(BC) $x^2 + y^2 - 8x - 16y + 60 = 0$

Equation of chord of contact from (–2, 0) is $-2x - 4(x-2) - 8y + 60 = 0$

$3x + 4y - 34 = 0 \qquad \ldots$ (ii)

From (i) and (ii)

$$x^2 + \left(\frac{34-3x}{4}\right)^2 - 8x - 16\left(\frac{34-3x}{4}\right) + 60 = 0$$

$16x^2 + 1156 - 204x + 9x^2 - 128x - 2176 + 192x + 960 = 0$

$5x^2 - 28x - 12 = 0 \quad \Rightarrow \quad (x-6)(5x+2) = 0$

$x = 6, -\frac{2}{5} \quad \therefore \quad$ Points are $(6, 4)\left(-\frac{2}{5}, \frac{44}{5}\right)$

10.(B) $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 12$

$1 \le x_i \le 12 \quad \Rightarrow \quad {}^{12-1}C_{6-1} = {}^{11}C_5$ and 1 way to be subtracted.

11.(CD) Let $z = x + iy$

From 2nd equation $x = 6$ put in (1)

$$3|(x-12)+yi| = 5|x+(y-8)i|$$

$$9[36+y^2] = 25[36+(y-8)^2] \quad \text{(substituting } x=6\text{)}$$

$$9\cdot 36 + 9y^2 = 25\cdot 36 + 25[y^2 + 64 - 16y]$$

$$16y^2 - 25\cdot 16y + 36\cdot 16 + 25\cdot 64 = 0$$

$$y^2 - 25y + 36 + 100 = 0$$

$$y^2 - 25y + 136 = 0; \qquad (y-17)(y-8) = 0$$

Then $y = 17$ or $y = 8 \Rightarrow$ (C), (D)

12.(AB) $\because \quad \sin C = \dfrac{1-\cos A\cos B}{\sin A\sin B} \le 1 \Rightarrow \cos(A-B) \ge 1$

$\Rightarrow \quad \cos(A-B) = 1 \quad \Rightarrow \quad A - B = 0 \Rightarrow A = B$

$\therefore \quad \sin C = \dfrac{1-\cos^2 A}{\sin^2 A} = 1 \Rightarrow C = 90°$

13.(4) We have $x, A_1, A_2, A_3, ...A_{19}, A_{20}, A_{21}, ...A_{99}, 2y$ are in A.P.

$\therefore$ Common difference of A.P. $= d = \dfrac{2y-x}{100}$

$$\Rightarrow \quad A_{20} = x + 20d = x + 20\left(\frac{2y-x}{100}\right) = x + \left(\frac{2y-x}{5}\right) = \frac{4x+2y}{5} \qquad \ldots (1)$$

$|||1y \quad 2x \le A'_1, A'_2, A'_3, ...A'_{19}, A'_{20}, A'_{21}, ...A'_{99}, y$ are in A.P.

$\therefore$ Common difference of above A.P. $= d' = \dfrac{y-2x}{100}$

$$\Rightarrow \quad A'_{20} = 2x + 20d' = 2x + 20\left(\frac{y-2x}{100}\right) = 2x + \left(\frac{y-2x}{5}\right) = \frac{8x+y}{5} \qquad \ldots (2)$$

Now $A_{20} = A'_{20}$ (Given)

$\therefore$ From (1) and (2), we get $\dfrac{4x+2y}{5} = \dfrac{8x+y}{5} \Rightarrow y = 4x$

14.(6) $T_{r+1} = {}^{10}C_r (1+3^{1/3})^r \cdot 7^{\frac{10-r}{7}}$

Clearly r can be 3 or 10

For $r = 3$

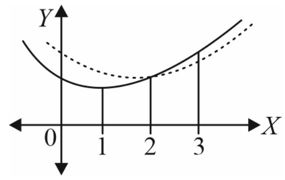
$${}^{10}C_3\left(1+3^{\frac{1}{3}}\right)^3 \cdot 7^1; \qquad {}^{10}C_3\,{}^3C_m\,3^{m/3}\cdot 7^1$$

Clearly m can be 0 to 3 = 2 terms.

For $r = 10$; ${}^{10}C_{10}\left(1+3^{1/3}\right)^{10} \cdot 7^0 \Rightarrow {}^{10}C_m\,3^{n/3}$

Clearly n can be 0, 3, 6, 9 $\equiv 4$ terms. Total terms $\equiv 6$.

15.(4) $P(x) = \underbrace{(m^2 + 4m + 5)}_{\text{always positive}} x^2 - 4x + 7$



Now, abscissa of vertex $= \dfrac{-b}{2a} = -\left(\dfrac{-4}{2(m^2 + 4m + 5)}\right) = \dfrac{2}{(m+2)^2 + 1}$

$\therefore$ Abscissa of vertex $= \dfrac{-b}{2a} \le 2$

Now, $3 \le x \le 5$, so

$P(x)|_{\min} = P(3)$

$= 9(m^2 + 4m + 5) - 12 + 7$

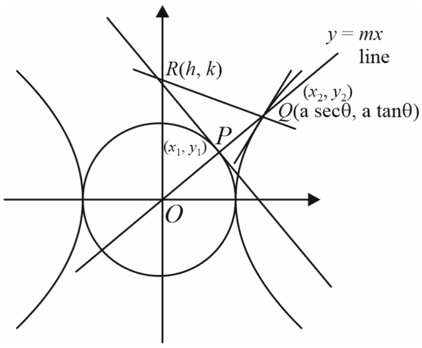
$= 9(m+2)^2 + 4$

$\therefore$ Minimum of minimum value of $P(x)$ is = 4 at m = – 2.

16.(4) Let the line by $y = mx$

Solving with the circle $x^2 + m^2x^2 = a^2$

$x_1 = \dfrac{a}{\sqrt{1+m^2}}$; $y_1 = \dfrac{am}{\sqrt{1+m^2}}$



Hence $P\left(\dfrac{a}{\sqrt{1+m^2}}, \dfrac{a\,m}{\sqrt{1+m^2}}\right)$

Solving with $x^2 - y^2 = a^2$

$x^2(1 - m^2) = a^2$

$x_2 = \dfrac{a}{\sqrt{1-m^2}}$; $y_2 = \dfrac{a\,m}{\sqrt{1-m^2}}$

Tangent at P

$$\frac{x \cdot a}{\sqrt{1+m^2}} + \frac{y \cdot am}{\sqrt{1+m^2}} = a^2$$

$x + my = a\sqrt{1+m^2}$; as it passes through $(h, k) \Rightarrow h + km = a\sqrt{1+m^2}$ … (1)

Tangent at Q

$$\frac{x \cdot a}{\sqrt{1-m^2}} - \frac{y \cdot am}{\sqrt{1-m^2}} = a^2$$

$x - my = a\sqrt{1-m^2}$; as it also passes through (h, k) hence $h - km = a\sqrt{1-m^2}$ … (2)

Squaring and adding as well as subtracting, we get

$h^2 = k^2m^2 = a^2$ and $4hk = 2a^2m \Rightarrow m = \dfrac{2hk}{a^2}$

$\therefore$ $h^2 + k^2 \cdot \dfrac{4h^2k^2}{a^4} = a^2 \Rightarrow$ result.

17.(2013) Let $\frac{\theta}{2^{n-1}} = A \quad \Rightarrow \quad \frac{2\theta}{2^n} = A \quad \Rightarrow \quad \frac{\theta}{2^n} = \frac{A}{2} \quad \Rightarrow \qquad \tan\left(\frac{\theta}{2^n}\right)\sec\left(\frac{\theta}{2^{n-1}}\right) = \tan\frac{A}{2} \cdot \sec A = \frac{\sin\frac{A}{2}}{\cos\frac{A}{2}.\cos A}$

$$= \frac{\sin\left(A - \frac{A}{2}\right)}{\cos\frac{A}{2} \cdot \cos A} = \frac{\sin A.\cos\frac{A}{2} - \cos A \cdot \sin\frac{A}{2}}{\cos\frac{A}{2} \cdot \cos A} = \tan A - \tan\frac{A}{2} \quad \Rightarrow \quad \tan\frac{\theta}{2^n} \cdot \sec\frac{\theta}{2^{n-1}} = \left(\tan\frac{\theta}{2^{n-1}} - \tan\frac{\theta}{2^n}\right)$$

$$\sum_{n=1}^{2013} \tan\frac{\theta}{2^n}\sec\frac{\theta}{2^{n-1}} = \sum_{n=1}^{2013}\left[\tan\left(\frac{\theta}{2^{n-1}}\right) - \tan\frac{\theta}{2^n}\right]$$

$$= \tan\left(\frac{\theta}{2^0}\right) - \tan\left(\frac{\theta}{2^1}\right) + \tan\left(\frac{\theta}{2^1}\right) - \tan\left(\frac{\theta}{2^2}\right) + \tan\left(\frac{\theta}{2^2}\right) - \tan\left(\frac{\theta}{2^3}\right) + \ldots + \tan\left(\frac{\theta}{2^{2012}}\right) - \tan\left(\frac{\theta}{2^{2013}}\right)$$

$$= \tan\left(\frac{\theta}{2^0}\right) - \tan\left(\frac{\theta}{2^{2013}}\right) \qquad \Rightarrow \qquad a = 0 \;\&\; b = 2013 \;\; \Rightarrow \;\; a + b = 2013.$$

18.(2) $(x-1)^2 + (y-2)^2 + \lambda(x - y + 1)^2 = 0$

Put $(1,0)$ to get λ.